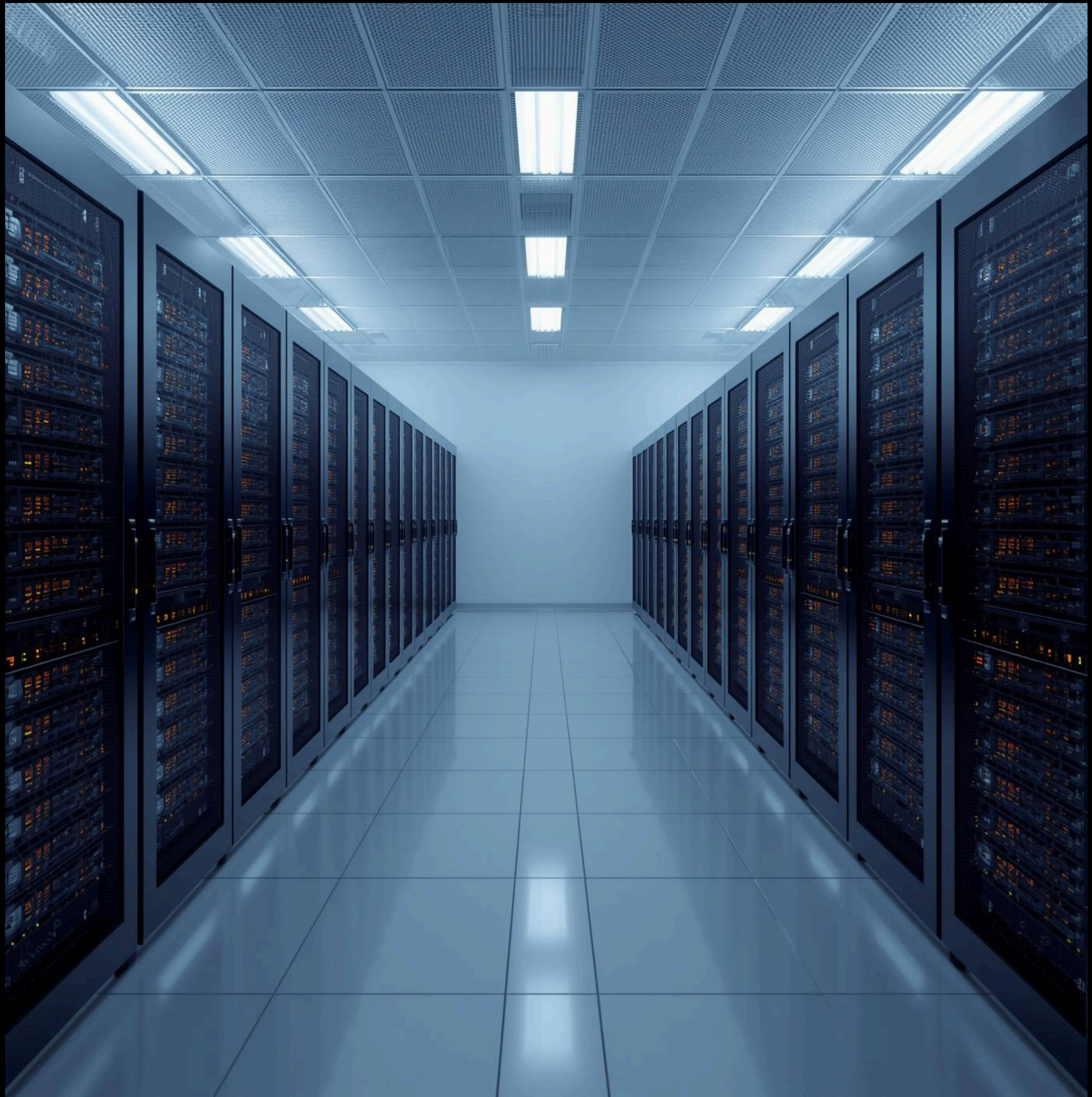
PROJECT PRESENTATION

Database Systems Project Presentation



Presented by: Group Oracle

Team Introduction

Ali Huseynov

Fronted / Backend Operations and
UI Views
Countries Table

Devrim Polat

Backend Operations,
Economy Indicators Table

Yunus Korkmaz

Database Management,
Health Indicators Table

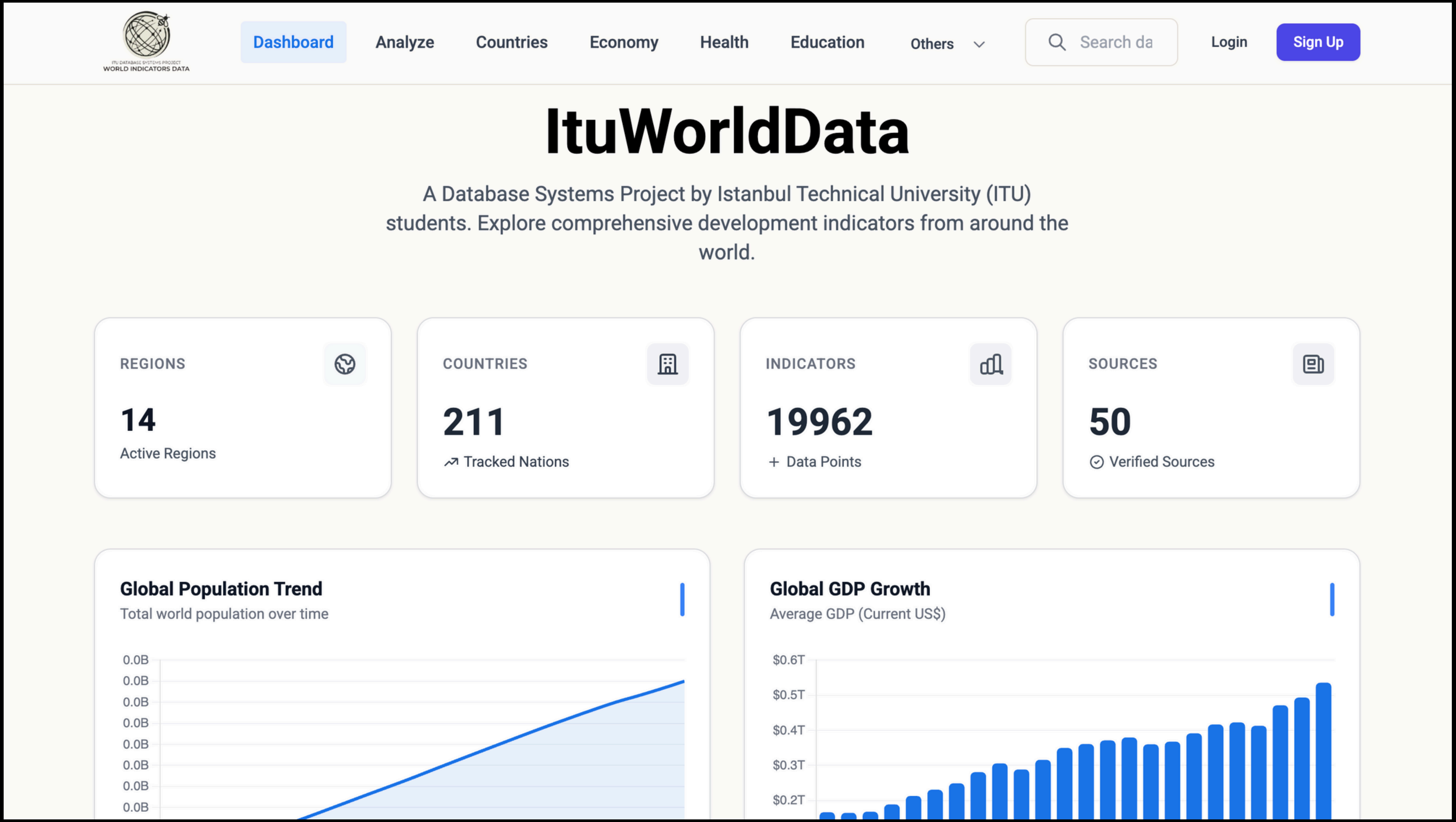
Ahmet Yusuf Kurukız

Backend Operations
Indicators Table

Mehmet Fatih Kaya

Frontend Operations and UI Views,
Education Indicators Table

Overview

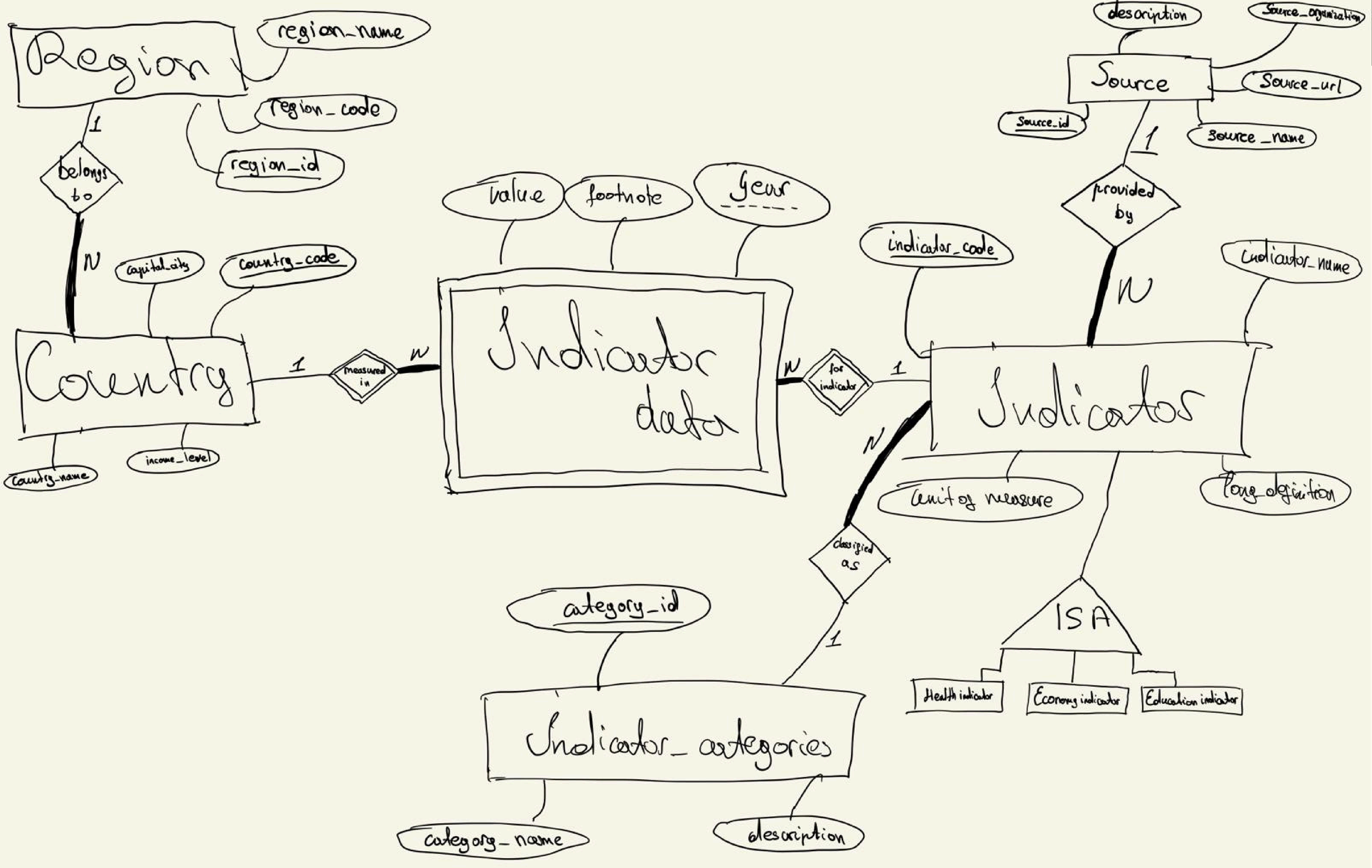

ITU DATABASE SYSTEMS PROJECT
WORLD INDICATORS DATA
Dashboard
Analyze
Countries
Economy
Health
Education
Others
Search da
Login
Sign Up
ItuWorldData
A Database Systems Project by Istanbul Technical University (ITU) students. Explore comprehensive development indicators from around the world.
REGIONS
14
Active Regions
COUNTRIES
211
Tracked Nations
INDICATORS
19962
Data Points
SOURCES
50
Verified Sources
Global Population Trend
Total world population over time
0.0B
0.0B
0.0B
0.0B
0.0B
0.0B
0.0B
0.0B
Global GDP Growth
Average GDP (Current US$)
$0.6T
$0.5T
$0.4T
$0.3T
$0.2T

Motivation and Goals

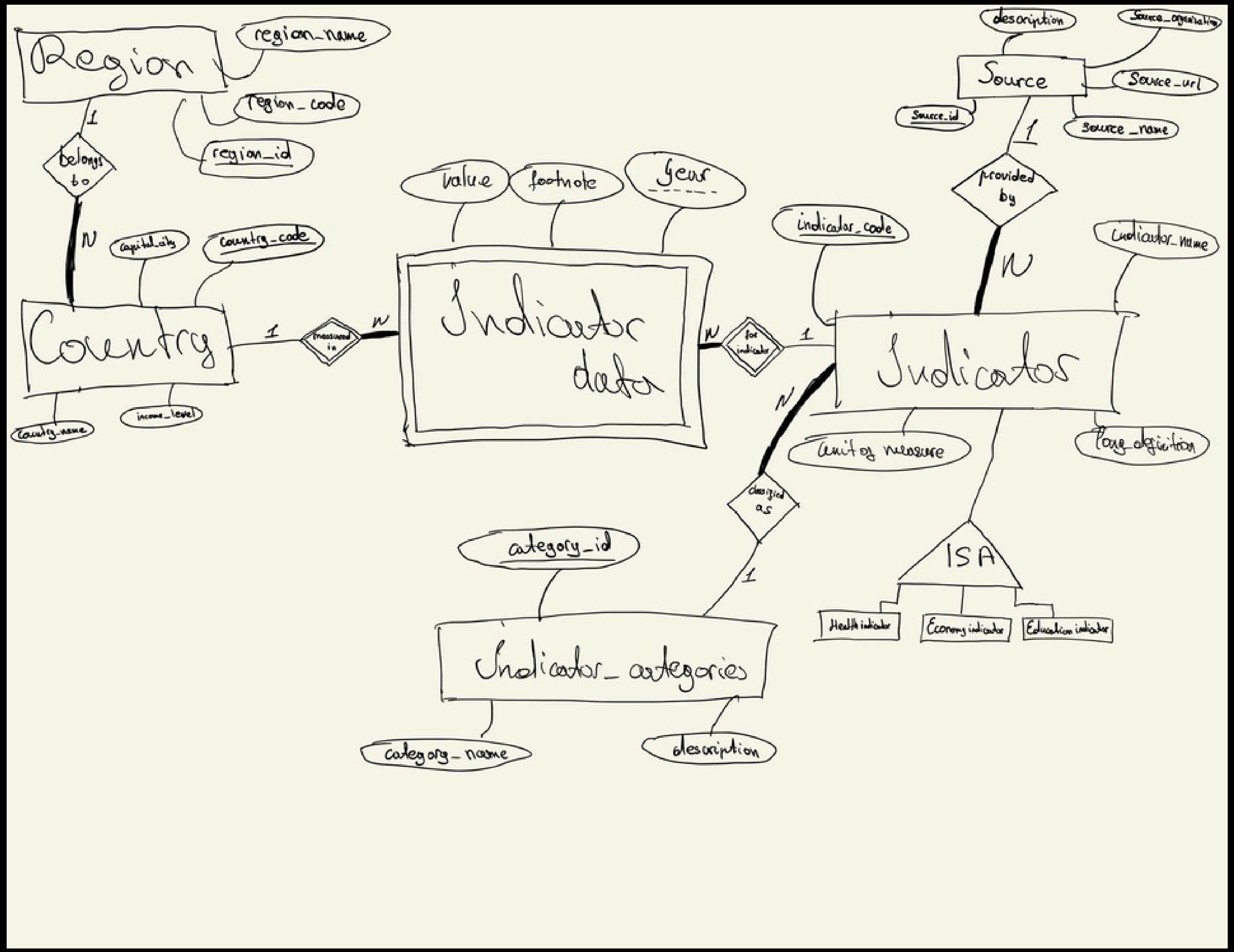
- **Data Fragmentation:** Critical global data is scattered across disparate sources (World Bank, WHO, UNESCO), making cross-domain analysis impossible
- **Redundancy:** Traditional flat-file storage leads to massive data duplication and update anomalies.

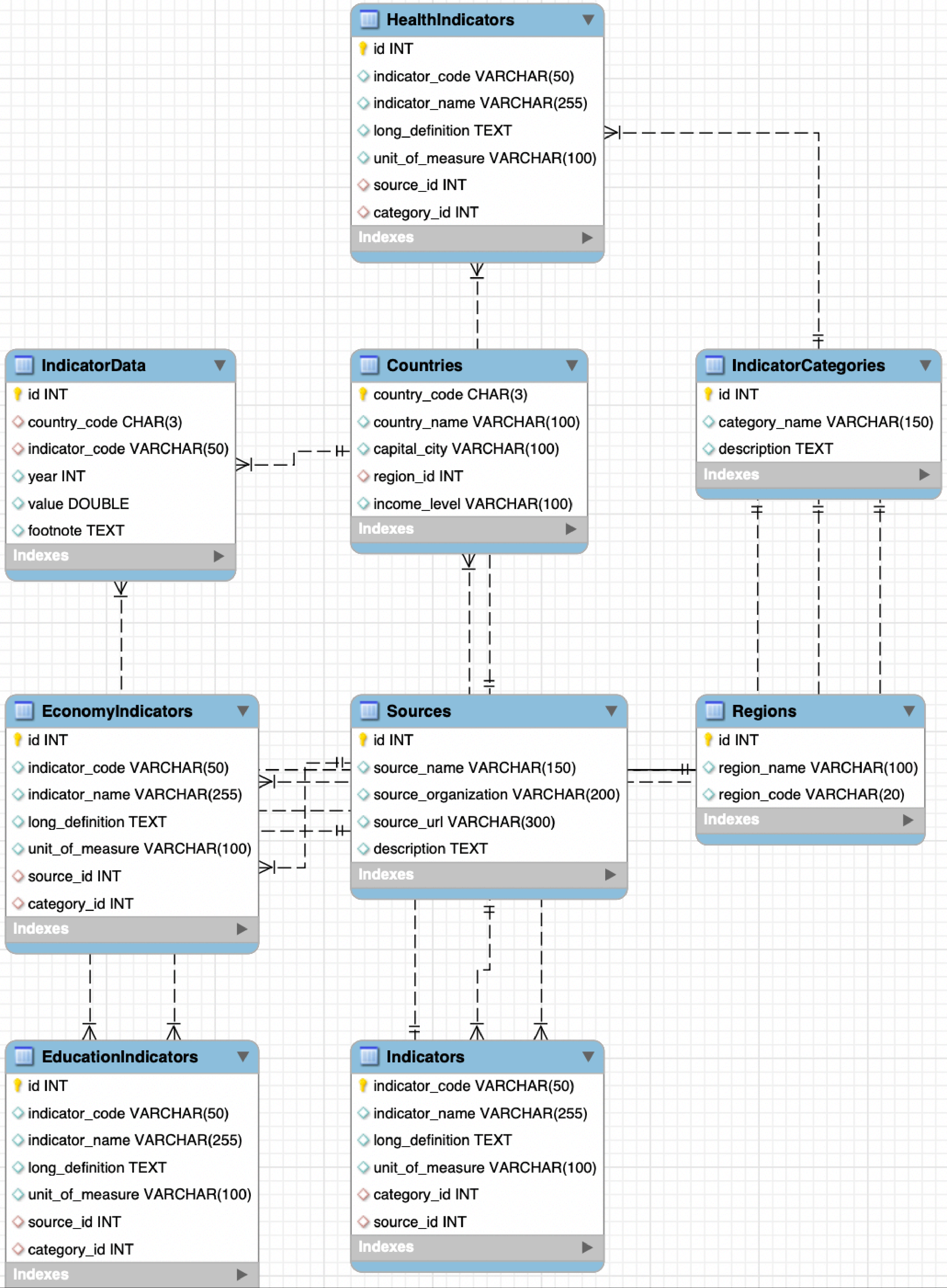
- **Normalization:** To design a 3NF schema that ensures data consistency
- **Security:** To implement a "Strict Admin Approval System" where new users must be explicitly approved by an Admin before accessing the data
- **Insight:** To transform raw static data into dynamic, comparative time-series visualizations

ER Diagram


Region
region_name
region_code
region_id
belongs to
Country
capital_city
country_code
country_name
income_level
measured in
Indicator data
value
footnote
year
for indicator
Indicator
indicator_code
indicator_name
unit of measure
long_definition
Source
description
source_organization
Source_url
Source_id
source_name
provided by
classified as
ISA
Health indicator
Economy indicator
Education indicator
category_id
Indicator_categories
category_name
description

ER-to-Relational Mapping:





Before Normalization

Before Normalization: Raw JSON

- Input: Raw API response in nested format.
- Redundancy: Source details are duplicated in thousands of rows.
- Inefficiency: High storage cost and risk of update anomalies.


```
t was reported by the donor to the Global Partnership for Education.","sourceOrganization":"The 2011 Monitoring Exercise on Aid Effectiv
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llocated and projected to specific developing countries. Data were later validated and\/or updated when preparing the GPE Results Report
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-effectiveness-2\/","topics":[]},{"id":"5.1.1_NER.TOTA.AID.AFD","name":"International aid disbursed to total education, AFD to Niger (US
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-effectiveness-2\/","topics":[]},{"id":"5.1.1_VNM.TOTA.AID.BEL","name":"International aid disbursed to total education, Belgium to Vietn
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```

Normalization

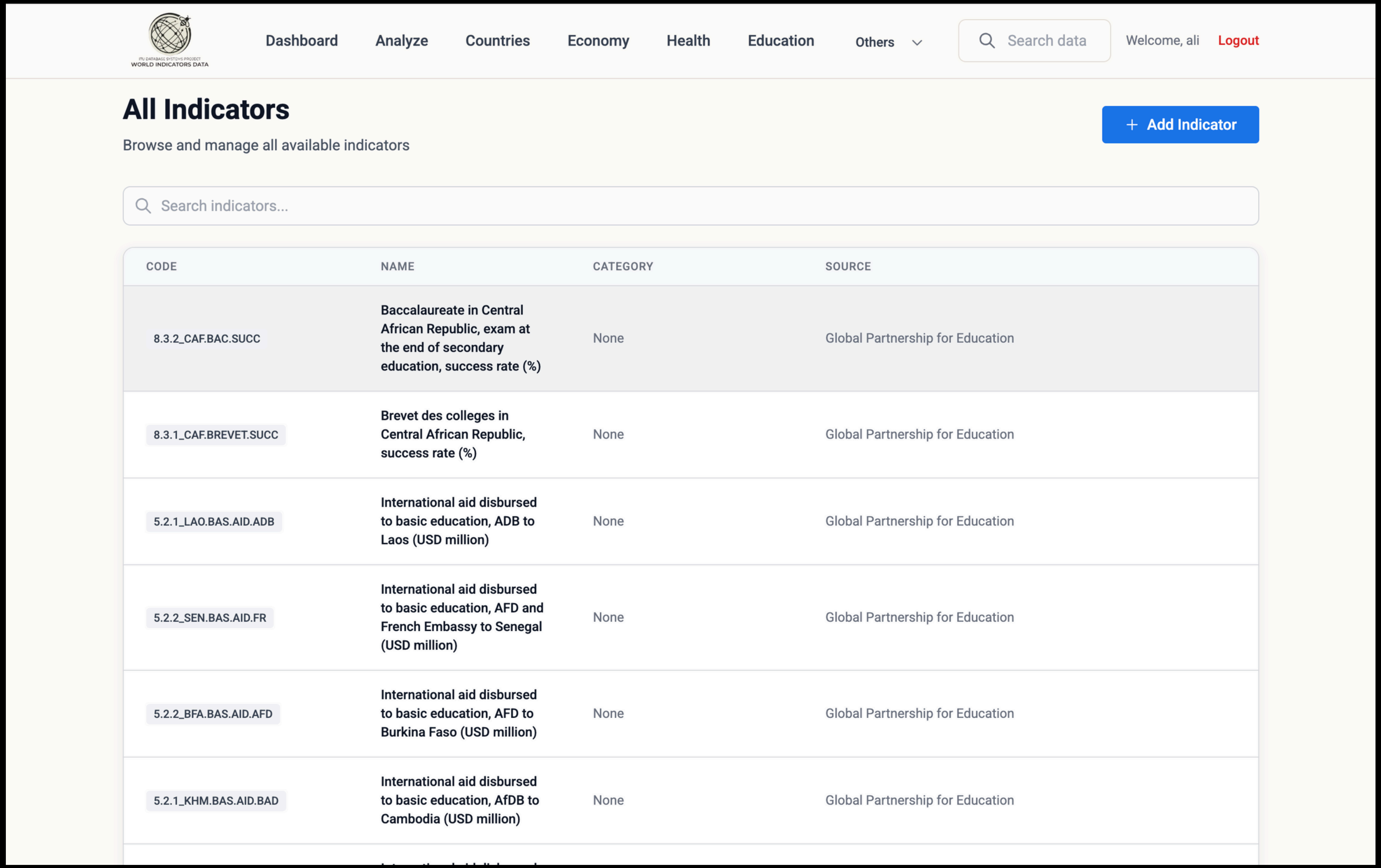
- ETL Approach: We used a Python script to Extract, Transform, and Load the data directly in 3NF structure.
- In-Memory De-duplication:
  - Utilized Python dictionaries (source_map) to identify and filter unique entries before database insertion.
  - Prevents duplicate records in Sources and Categories tables.
- Foreign Key Association:
  - The script retrieves the unique ID of the source/category.
  - Inserts only this Foreign Key (sid) into the main Indicators table, ensuring data integrity.

```python
# source
src = ind.get("source", {})
sid = src.get("id")

cursor.execute(sql, (
    code,
    name,
    long_def,
    unit,
    tid,
    sid
))
```

```python
# Collect sources
source_map = {}
for ind in indicators:
    src = ind.get("source", {})
    sid = src.get("id")
    if sid:
        source_map[sid] = {
            "name": src.get("value"),
            "organization": src.get("organization"),
            "url": src.get("url")
        }
print(f"Unique sources: {len(source_map)}")
```

After Normalization

- Structured Data: The complex, nested JSON objects are now stored in relational tables (Indicators, Sources, Categories).
- Reference-Based Storage: The "Source" column is no longer a repeated text block. It is a Foreign Key reference to the Sources table.
- 3NF Compliance:
- Transitive Dependencies Removed: Source details (organization name, description) are separated from the Indicator data.
- Update Anomalies Eliminated: Updating a source's name in the Sources table automatically updates it for all associated indicators.



Dataset Description

What does the data include?: The dataset comprises time-series development metrics categorized into three core domains: Economy, Health, and Education. It covers global data for countries and regions, including variables such as GDP, Population, Life Expectancy, and School Enrollment rates.

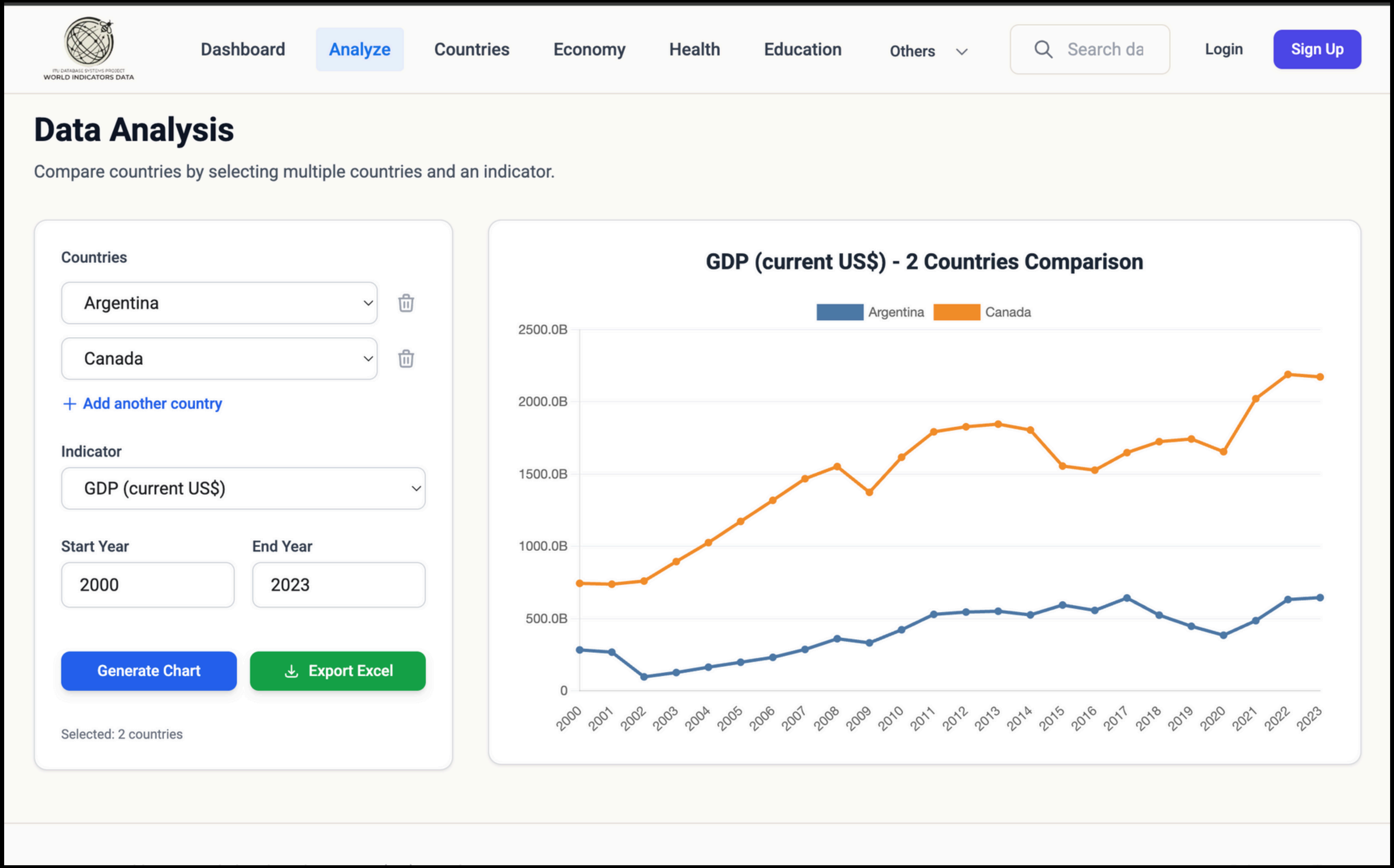
What is the source?: The primary source is the World Development Indicators (WDI) database managed by the World Bank. We utilized the official WDI API and bulk CSV exports to ensure data accuracy and reliability.

How large is it?: The database contains 19962 rows, covering information about 211 countries.

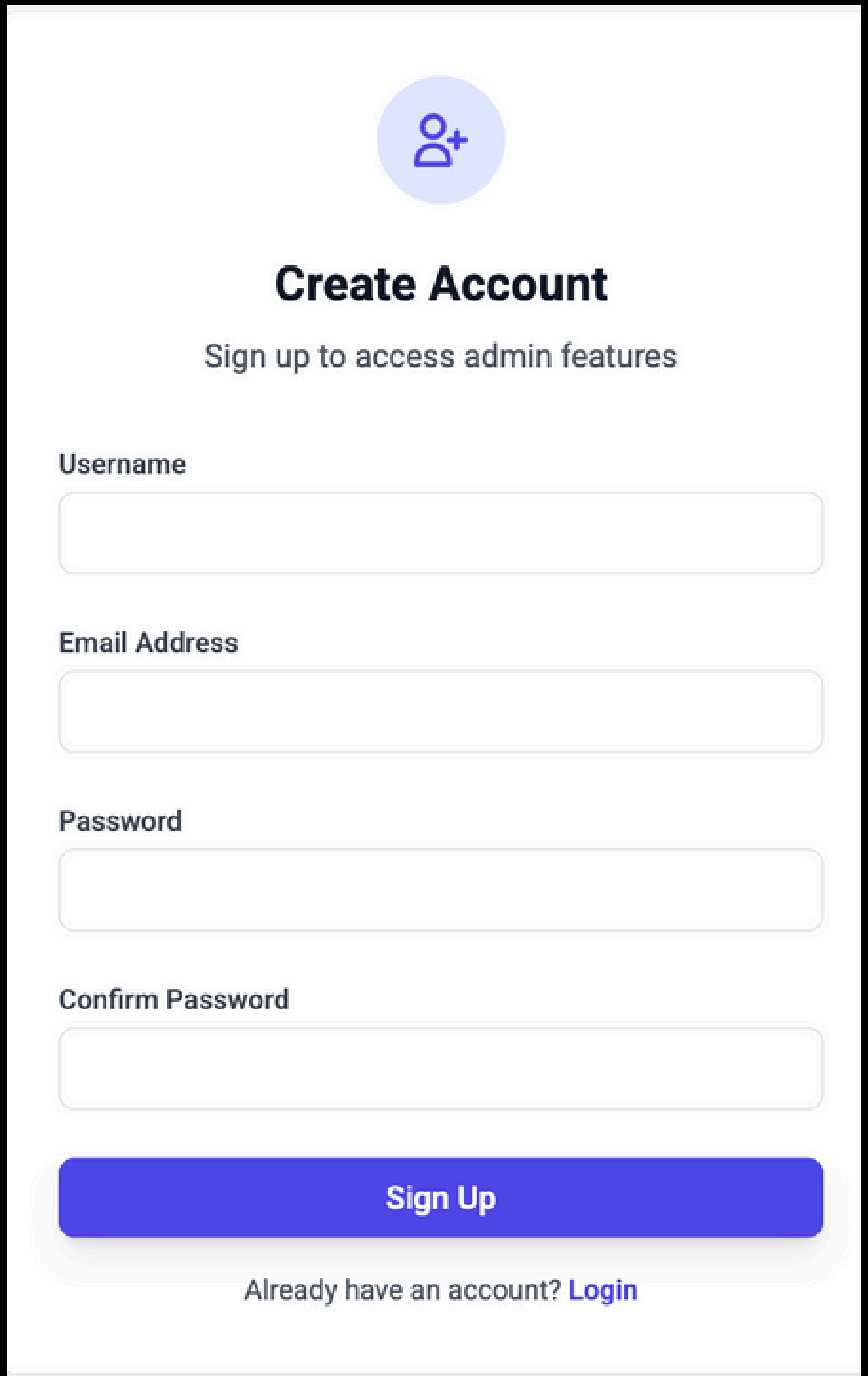
Data Fetching

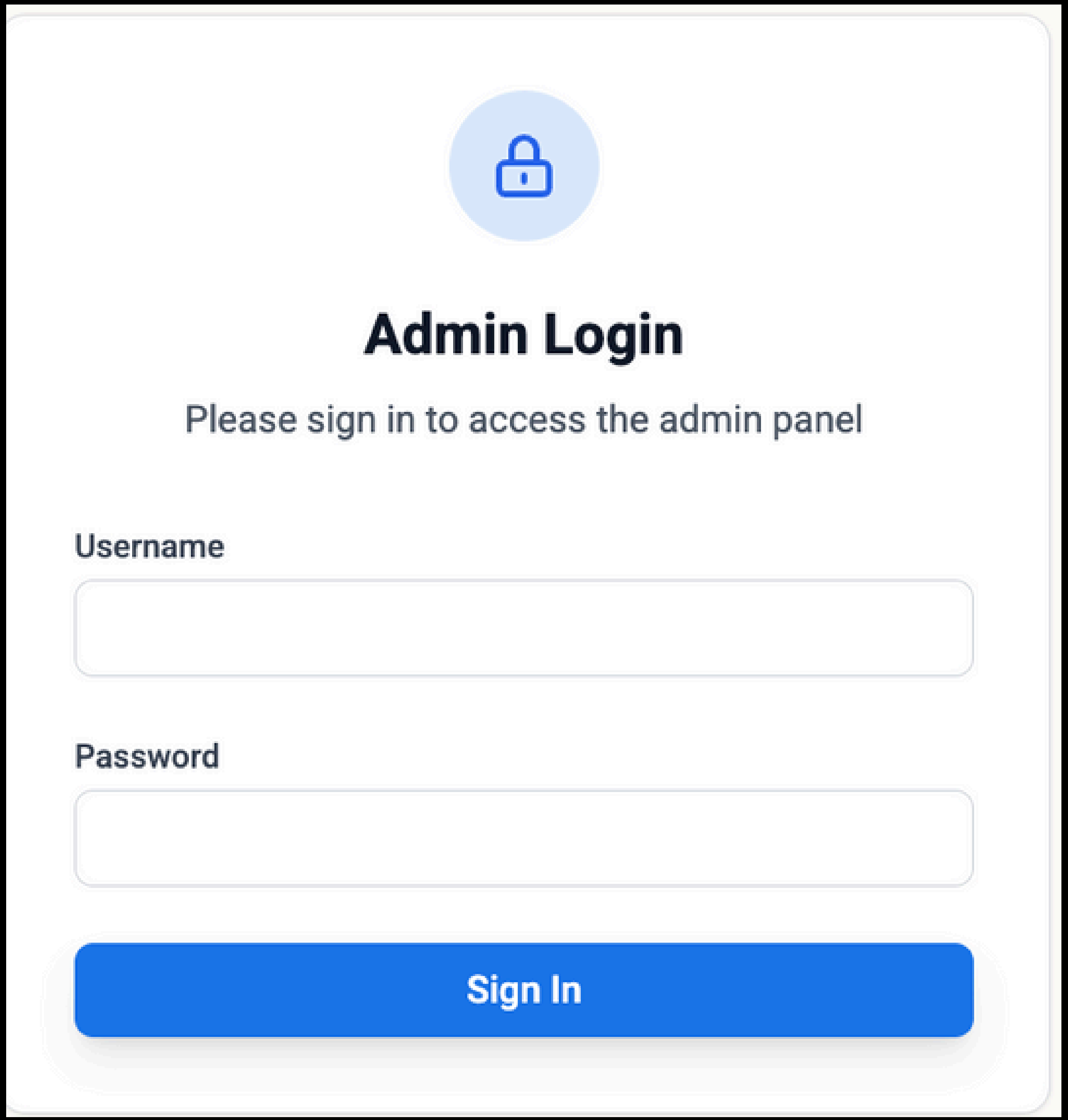
```python
def fetch_indicator_all(indicator_code):
    url = (
        f"https://api.worldbank.org/v2/country/all/indicator/"
        f"{indicator_code}?format=json&per_page=20000"
    )

    for attempt in range(5):
        resp = requests.get(url)
        if resp.status_code == 200:
            data = resp.json()
            if len(data) < 2 or data[1] is None:
                return []
            return data[1]

        if resp.status_code in (500, 502, 503, 504):
            print(f"[{indicator_code}] Server {resp.status_code}, retry {attempt+1}/5")
            time.sleep(1)
            continue

        resp.raise_for_status()

    print(f"[{indicator_code}] SKIPPED: API unreachable.")
    return []
```

Compare Countries



Sign Up and Log In

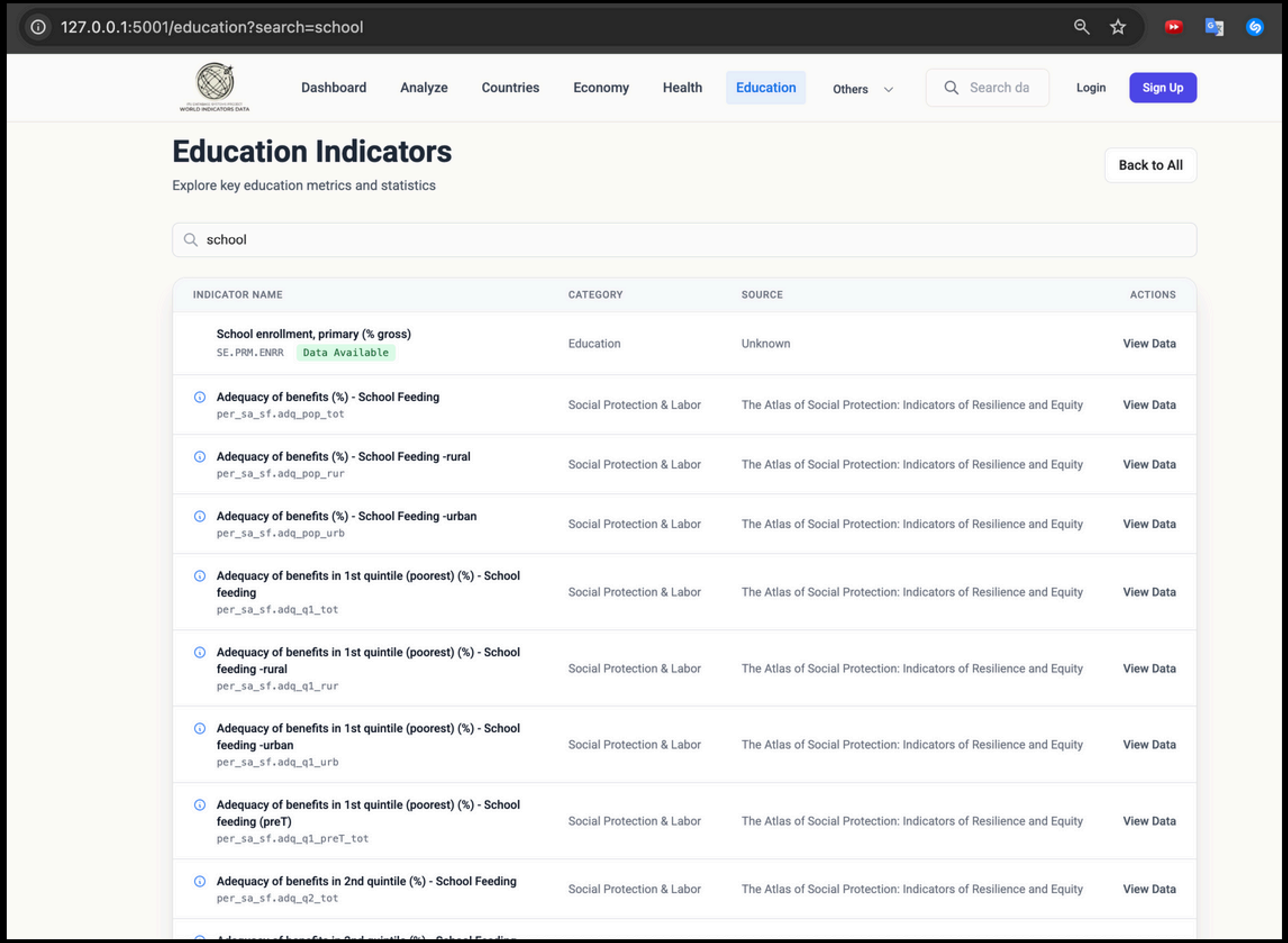




```python
def init_db():
    conn = get_db_connection()
    cursor = conn.cursor()

    # Create Users table
    with open('database/create_users_table.sql', 'r') as f:
        cursor.execute(f.read())

    # Check if admin exists
    cursor.execute("SELECT * FROM Users WHERE username = 'huseynov'")
    if not cursor.fetchone():
        # Create admin user
        password = "09052006"
        password_hash = generate_password_hash(password, method='pbkdf2:sha256')
        cursor.execute(
            "INSERT INTO Users (username, password_hash, role) VALUES (%s, %s, %s)",
            ('huseynov', password_hash, 'admin')
        )
        print("Admin user 'huseynov' created.")
    else:
        print("Admin user already exists.")

    conn.commit()
    cursor.close()
    conn.close()

if __name__ == '__main__':
    init_db()
```

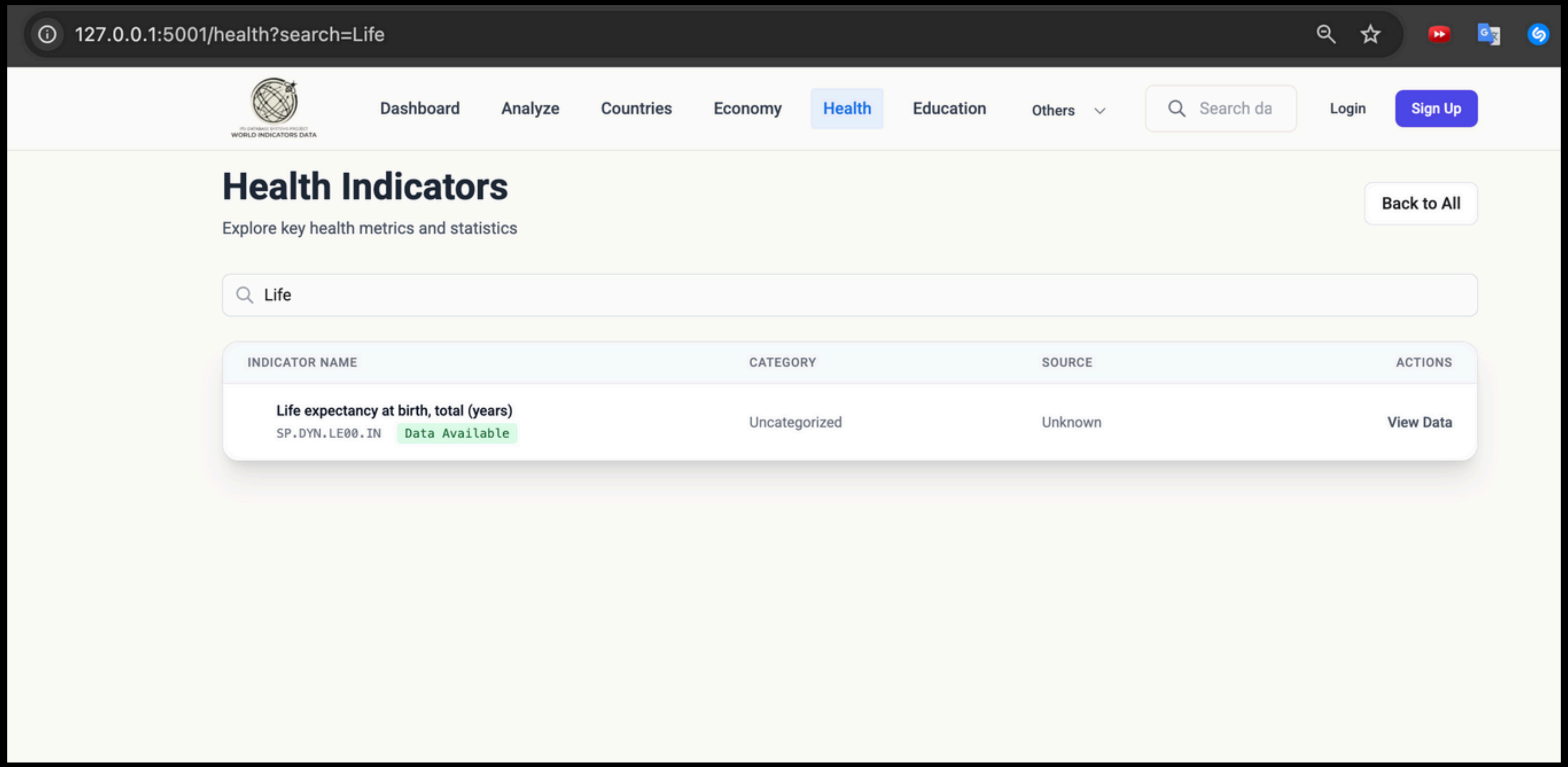
Complex Queries

```python
def get_education_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
            SELECT COUNT(*) as count
            FROM EducationIndicators i
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data,
            (CASE WHEN i.long_definition IS NOT NULL AND i.long_definition != '' THEN 1 ELSE 0 END) as is_well_defined
            FROM EducationIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            ORDER BY has_data DESC,is_well_defined DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM EducationIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data,
            (CASE WHEN i.long_definition IS NOT NULL AND i.long_definition != '' THEN 1 ELSE 0 END) as is_well_defined
            FROM EducationIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            ORDER BY has_data DESC,is_well_defined DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, fetch=True)

    return items, total_count
```



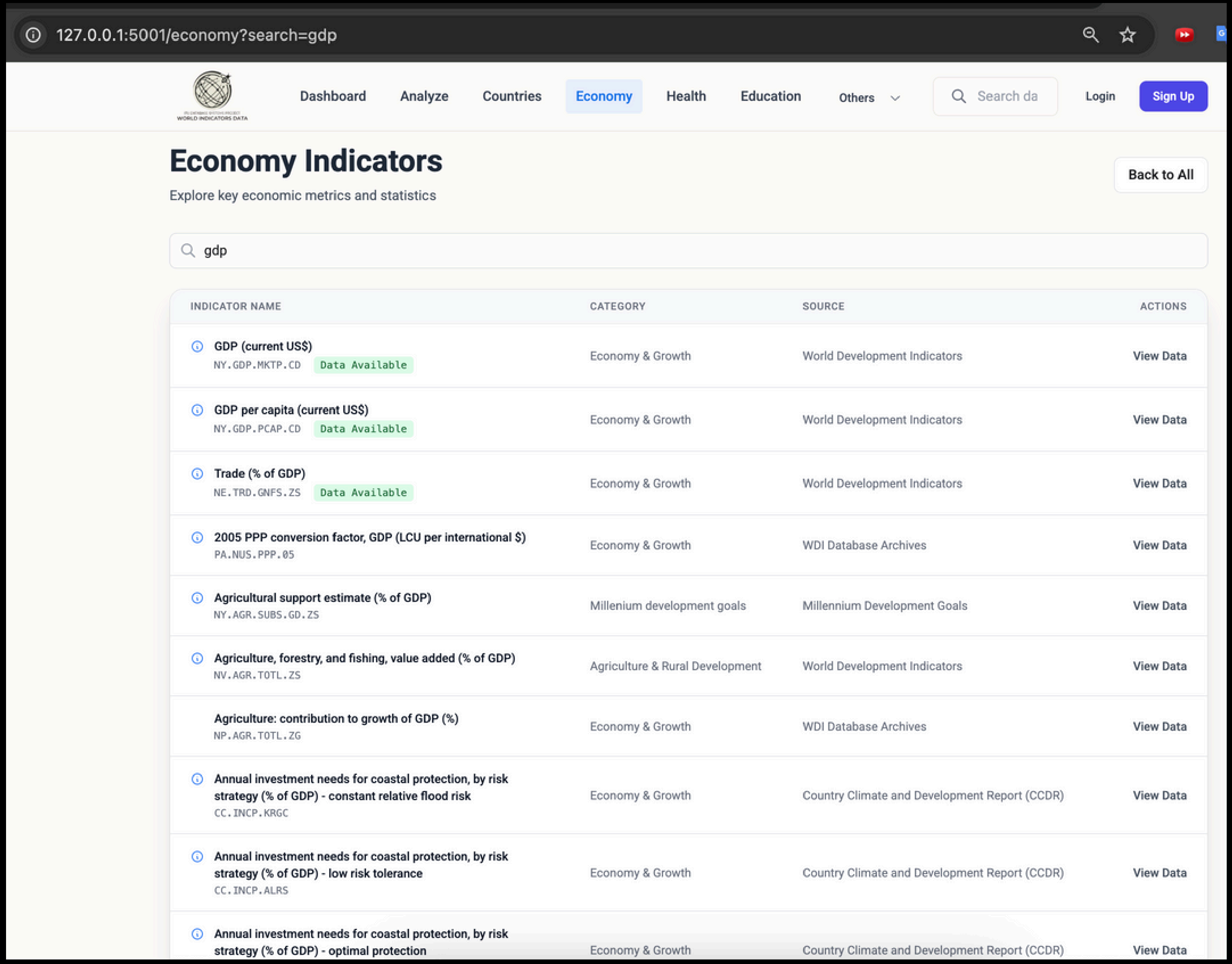
Complex Queries

```python
def get_health_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
            SELECT COUNT(*) as count
            FROM HealthIndicators i
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM HealthIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM HealthIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            EXISTS(SELECT 1 FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM HealthIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, fetch=True)

    return items, total_count
```



Complex Queries

```python
def get_economy_indicators(page=1, per_page=1000, search_query=None):
    offset = (page - 1) * per_page

    if search_query:
        count_query = """
            SELECT COUNT(*) as count
            FROM EconomyIndicators i
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
        """
        count_result = execute(count_query, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            (SELECT COUNT(*) FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM EconomyIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            WHERE i.indicator_name LIKE %s OR i.indicator_code LIKE %s
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, (f"%{search_query}%", f"%{search_query}%"), fetch=True)
    else:
        count_result = execute("SELECT COUNT(*) as count FROM EconomyIndicators", fetch=True)
        total_count = count_result[0]['count'] if count_result else 0

        items = execute(f"""
            SELECT i.*, s.source_name, c.category_name,
            (SELECT COUNT(*) FROM IndicatorData d WHERE d.indicator_code = i.indicator_code) as has_data
            FROM EconomyIndicators i
            LEFT JOIN Sources s ON i.source_id = s.id
            LEFT JOIN IndicatorCategories c ON i.category_id = c.id
            ORDER BY has_data DESC, i.indicator_name ASC
            LIMIT {per_page} OFFSET {offset}
        """, fetch=True)

    return items, total_count
```



Complex Queries

```python
def get_indicators_with_data():
    """Get only indicators that have actual data in IndicatorData table"""
    return execute(
        """
        SELECT DISTINCT I.indicator_code, I.indicator_name, I.source_id, I.category_id, I.long_definition,
               C.category_name, S.source_name
        FROM Indicators I
        INNER JOIN IndicatorData D ON I.indicator_code = D.indicator_code
        LEFT JOIN IndicatorCategories C ON I.category_id = C.id
        LEFT JOIN Sources S ON I.source_id = S.id
        ORDER BY I.indicator_name
        """,
        fetch=True
    )
```

Complex Queries

```python
def get_yearly_indicator_average(indicator_code):
    return execute(
        """
        SELECT year, AVG(value) AS avg_value
        FROM IndicatorData
        WHERE indicator_code=%s
        GROUP BY year
        ORDER BY year
        """,
        (indicator_code,),
        fetch=True
    )

def get_global_population():
    return get_yearly_indicator_average("SP.POP.TOTL")
```

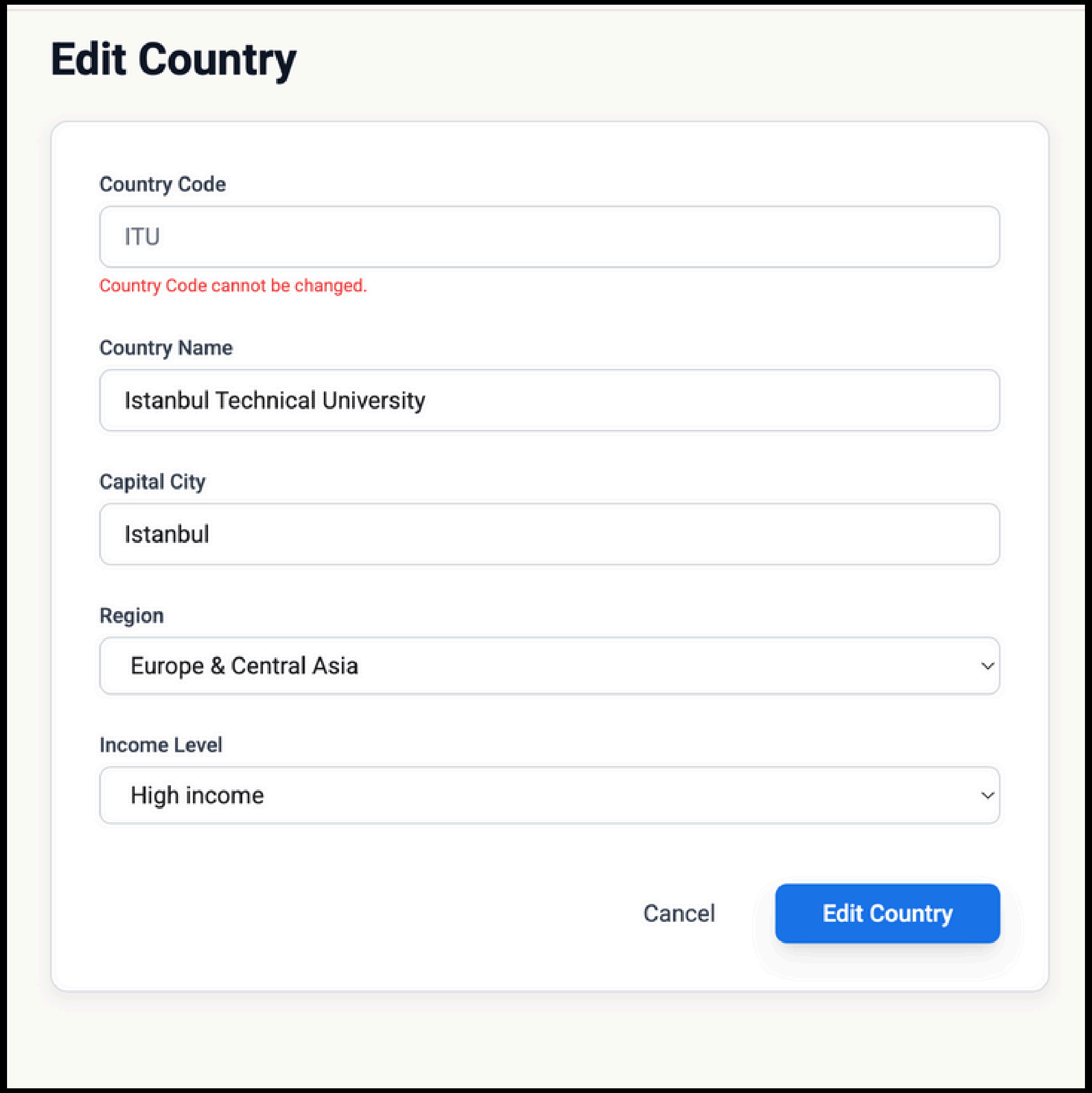
Complex Queries

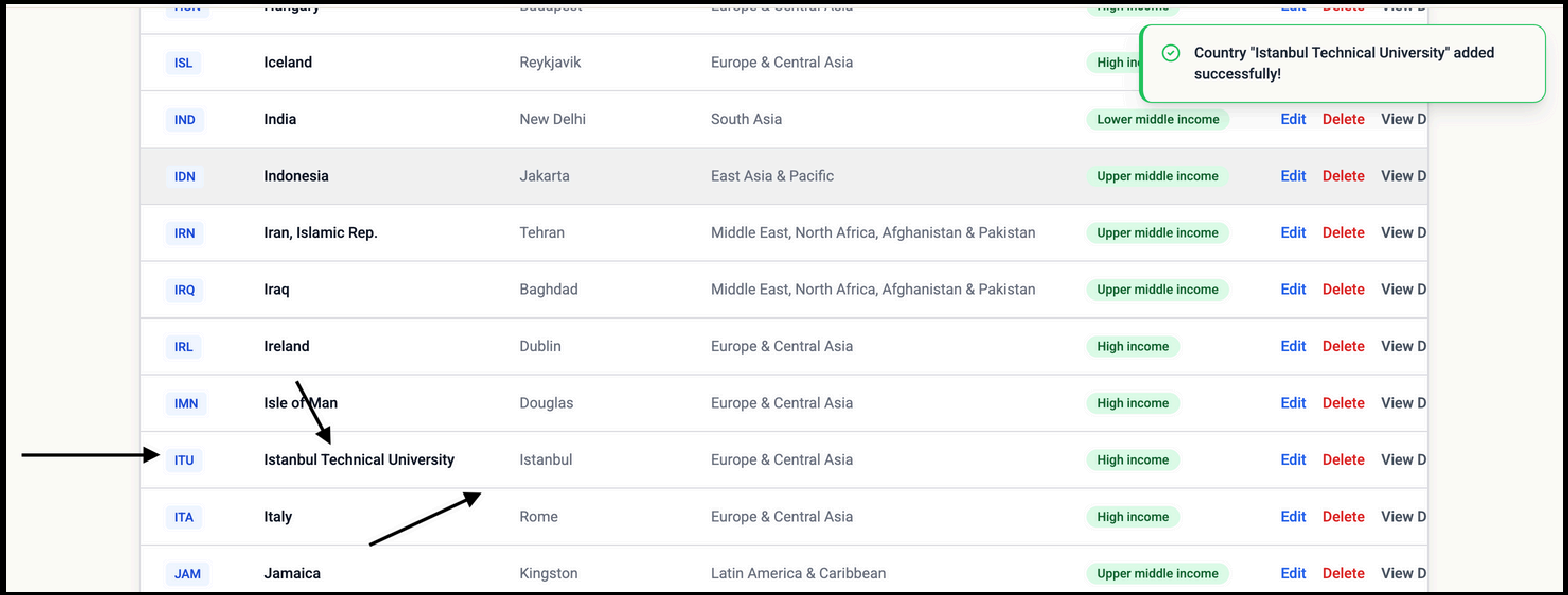
```python
def get_yearly_indicator_average(indicator_code):
    return execute(
        """
        SELECT year, AVG(value) AS avg_value
        FROM IndicatorData
        WHERE indicator_code=%s
        GROUP BY year
        ORDER BY year
        """,
        (indicator_code,),
        fetch=True
    )

def get_global_population():
    return get_yearly_indicator_average("SP.POP.TOTL")
```

Delete Constraints

- **IndicatorData Table :**
  - **Countries:** ON DELETE RESTRICT
  - **Indicators:** ON DELETE RESTRICT
- **EducationIndicators Table :**
  - **IndicatorCategories:** ON DELETE CASCADE
  - **Sources:** ON DELETE RESTRICT
- **EconomyIndicators Table :**
  - **IndicatorCategories:** ON DELETE RESTRICT
  - **Sources:** ON DELETE RESTRICT
- **Health Indicators Table :**
  - **IndicatorCategories:** ON DELETE CASCADE
  - **Sources:** ON DELETE RESTRICT
- **Countries Table :**
  - **Regions:** ON DELETE RESTRICT

Testing





"Since this is a data-driven application with limited user interaction, our testing mainly focused on validating imported data, enforcing admin-only CRUD operations, and ensuring database integrity."

Thank you for your attention!